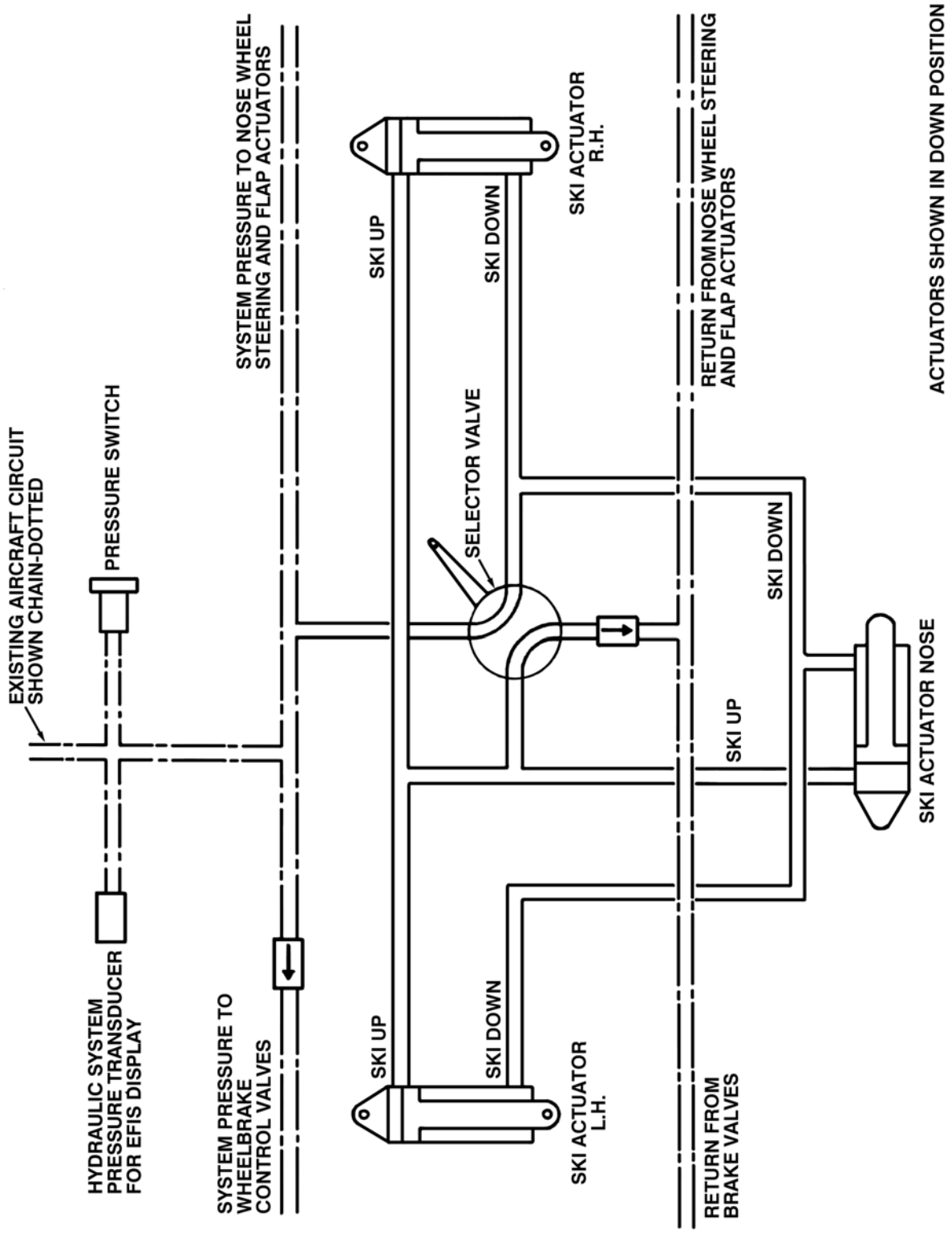
Extension and Retraction (Wheel-Skis) (Mod S.O.O. 6001) – General data

1. General

Refer to Figure 1 for details.

Extension and retraction of the wheel-skis is accomplished by directing hydraulic fluid, under pressure from the aircraft's hydraulic system (refer to 29-00-00, Hydraulic Power – General Data), to either the “up” or “down” side of the wheel-ski hydraulic actuators. The direction of system fluid flow, and the return of the displaced fluid, is determined by the position of a four-way selector valve. The selector valve, mounted behind the ski position selector valve on the instrument panel. The ski position selector valve has two positions marked UP/LAND and DOWN/SNOW. The panel also includes indicator lights to provide ski position indication.



Wheel-Ski Hydraulic System – Schematic
Figure 1

Print Date: 2025-09-15

2. Adjustment/Test

A. Rig Wheel-Ski Selector Control

Refer to Figure 2 for details.

1) Install Wheel-Ski position selector control in instrument panel cutout on the right hand side of the flight compartment.

2) Connect hydraulic hoses to ski position valve using straight/inline fittings.

3) If installed, remove caps from hydraulic bulkhead fittings for pressure lines labeled 'UP', 'DN', 'P' and 'R'.

4) From ski position valve, connect hydraulic hoses to bulkhead fittings labeled 'UP', 'DN', 'P' and 'R' using the 90° fittings.

5) Function test Wheel-Ski system, refer to Para 2.B.

B. Function Test Wheel-Ski Installation

1) Jack aircraft clear of ground. Use the main inboard jacking points and a narrow base jack (or jacking beam SD12538) to jack the nose (refer to 07-00-00, Lifting and Shoring – General Data).

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Ensure GEAR IND circuit breaker is engaged.

4) If the electrical hydraulic pump is to be used, engage the HYDRAULIC OIL PUMP circuit breaker.

Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi.

5) If the electrical hydraulic pump is not to be used, operate the hydraulic handpump.

6) Select ski position selector lever “UP/LAND” and “DOWN/SNOW” several times and check SKI POSITION indicators display three lights when at UP, and three when at DOWN, for the appropriate selection.

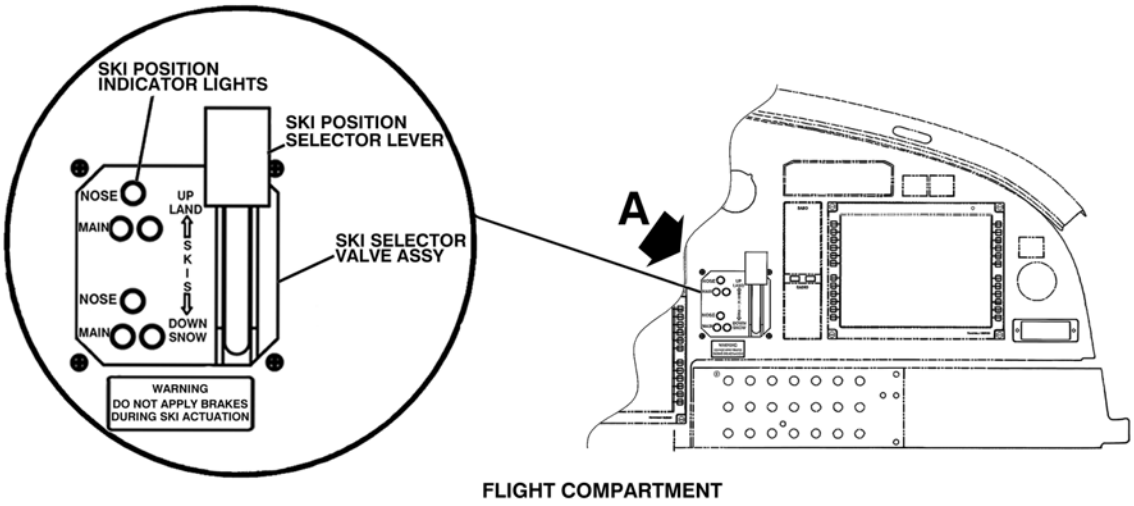
Note During function test, a hydraulic low pressure caution or warning CAS message might be generated, if so do steps Para 2.B.7) and Para 2.B.8).

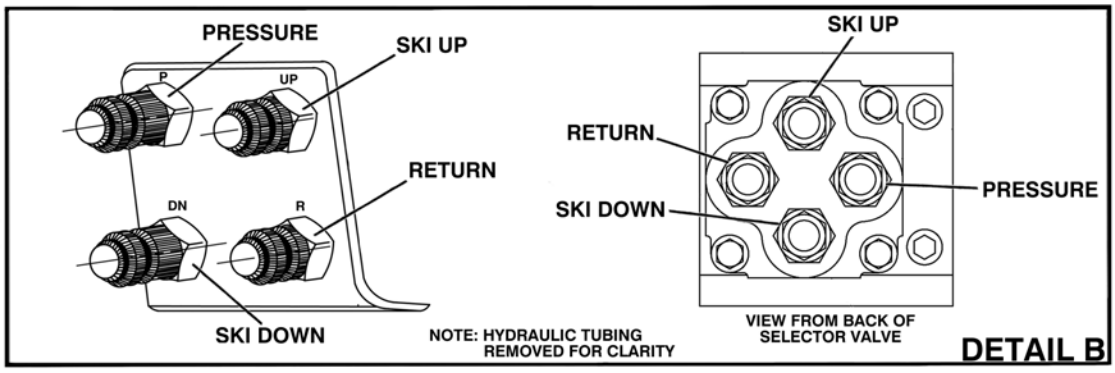
7) Confirm when skis are full “UP” or full “DOWN” the CAS message clears.

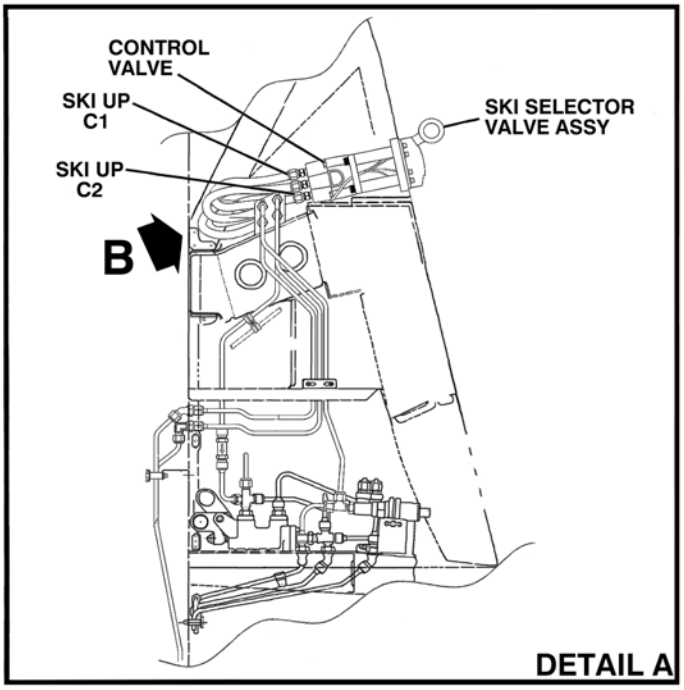
8) Check hydraulic system for leaks. If none, system is acceptable.

9) Lower aircraft to ground in the “UP/LAND” configuration, and remove jacking equipment.

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).







Ski Selector - Rigging
Figure 2